Summary of Hypothesis Testing & Sampling

POLI 380

1 Key Concepts in Hypothesis Testing

Hypothesis testing is the process of evaluating whether patterns in data support or contradict a specific claim. The framework involves:

- **Null Hypothesis ($\mathbf{H_0}$)**: Statement that the relationship of interest does not exist
- **Alternative Hypothesis ($\mathbf{H_a}$)**: Statement that the relationship of interest does exist
- **Statistical Inference**: Process of deciding whether to reject the null based on the data

The fundamental question in hypothesis testing is: *How unlikely is it that the results we find in our data could have emerged purely due to chance, assuming the null is true?*

2 Types of Errors in Hypothesis Testing

When making inferential decisions based on sample data, two types of errors are possible:

Inferential Decision	**Is H_0 True or False?**	
Based on Sample Data	**H_0 is true**	**H_0 is false**
Do not reject H_0	Correct Inference	Type II Error
Reject H_0	Type I Error	Correct Inference

Table 1: Type I and Type II Errors in Hypothesis Testing

This parallels the criminal justice system, where similar trade-offs exist:

Jury's Decision	**Is the Defendant Innocent or Guilty?**	
Based on Evidence	**Innocent**	**Guilty**
Acquit	Correct decision	Release guilty person
Convict	Convict innocent person	Correct decision

Table 2: Parallels Between Hypothesis Testing and Criminal Justice

3 Statistical Significance

- **p-value**: The probability of obtaining results at least as extreme as those observed, assuming the null hypothesis is true
- **Statistical significance threshold**: Conventionally set at $\alpha = 0.05$ (5%)
- **Interpreting significance**:
  - If p-value ¡ 0.05: Result is statistically significant, reject H_0
  - If p-value 0.05: Result is not statistically significant, fail to reject H_0

Important: Statistical significance does not necessarily imply substantive importance.

Effect of Treatment on Outcome	Substantive Effect Size	
	Weak	Strong
Statistically insignificant	Negligible concern	Possible Type II error
Statistically significant	Real but may not matter	Important finding

Table 3: Statistical vs. Substantive Significance

4 Sampling Methods and Evolution

4.1 Historical Development

Public opinion polling has evolved through several methodological stages:

- **1936 Literary Digest Poll**: Used a non-representative sample from telephone directories and auto registrations, incorrectly predicting Landon would defeat Roosevelt
- **Quota Sampling**: Improved representativeness by matching sample demographics to population characteristics
- **Probability Sampling**: Each unit has an equal chance of selection, allowing for proper statistical inference

4.2 Sources of Sampling Error

Sampling error can be expressed as:

$$\text{Population Parameter} = \text{Sample Statistic} + \text{Random Sampling Error} + \text{Systematic Sampling Error} \tag{1}$$

Where:

- **Random Sampling Error**: Natural variation due to chance in the sampling process
- **Systematic Sampling Error**: Biases in how units are sampled from the population

Random sampling error can be further defined as:

$$\text{Random Sampling Error} = \frac{\text{Variation Component}}{\text{Sample Size Component}} \tag{2}$$

This explains why:

- Higher variation in the population increases sampling error
- Larger sample size decreases sampling error

5 Confidence Intervals

Confidence intervals quantify the uncertainty around an estimate due to random sampling error:

$$\text{Confidence Interval} = \text{Estimate} \pm \text{Margin of Error} \tag{3}$$

Where the margin of error depends on:

- Sample size (larger samples = smaller margins)
- Confidence level (higher confidence = wider margins)
- Population variation (more variation = wider margins)

6 Political Polling Examples

Historical examples demonstrate both successes and failures in political polling:

- **2008 Presidential Polls**: Were generally accurate in predicting Obama's victory
- **2016, 2020, and 2024 Elections**: Polls consistently underestimated Republican support

Reasons for polling errors include:

- Changing demographics not captured in sampling frames
- Differential non-response across political groups
- Shifts in voter sentiment close to election day
- Challenges in identifying likely voters

7 Key Takeaways

1. Research is about rejecting the null hypothesis through rigorous testing
2. Representativeness of samples is more important than sample size
3. All sample-based inferences contain some uncertainty
4. Statistical significance does not guarantee substantive importance
5. Both measurement error and sampling error affect the quality of evidence